Appendix A
Influence of our trading strategy on the market

In order to better understand the potential for our trading strategy to influence the strategy of other agents, we have compared the quantity that we exchange for different delivery hours with the total exchanged quantity in the market. More precisely, we have recorded the total exchanged quantity in the German CIM for every hour of 10 days that are randomly selected in our dataset. We have then computed the ratio between our traded quantity and the total exchanged quantity for these hours. We present the results graphically in figure 1. It can be observed that, for most hours, the ratio is quite low (the average ratio is around 1.65%). For these hours, it could be argued that our influence on the future behavior of other actors could be negligible. On the contrary, for a few hours, our trading volume can reach a ratio of 10%.

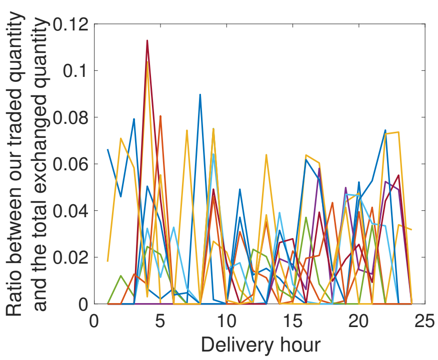

Fig. 1: Ratio between our traded quantity and the total exchanged quantity for 10 randomly selected days.

Appendix B
Threshold policy for a three stage stochastic program

In this appendix, we prove that a threshold policy is optimal for a three stage stochastic program. We consider the case in which we have a complete scenario tree which splits into N_1 scenarios at the second stage, and $N_1 \cdot N_2$ scenarios at the third stage. In order to describe the model, we need to define the sets, the parameters and the variables.

- The sets are:
  - D, the set of delivery times.
  - $S_1 = \{1..N_1\}$, the set of scenario index for the second stage.
  - $S_2 = \{1..N_2\}$, the set of scenario index for the third stage.
  - $I_{1,d}, \forall d \in D$, the set of bids that are available at the first time step and have a delivery time d.
  - $I^j_{2,d}, \forall d \in D, j \in S_1$, the set of bids that are available at the second step for the scenario j and have a delivery time d.
  - $I^{j,k}_{3,d}, \forall d \in D, j \in S_1, k \in S_2$, the set of bids that are available at the third step for the scenario j, k and have a delivery time d.
- The parameters are:
  - p^b_i, the price of the i^{th} buy bid.
  - p^s_i, the price of the i^{th} sell bid.
  - Q^b_i, the available quantity of the i^{th} buy bid.
  - Q^s_i, the available quantity of the i^{th} sell bid.
  - V is the maximum energy capacity of the reservoir.
- The variables are:
  - q^b_i, the quantity that we accept from the i^{th} buy bid.
  - q^s_i, the quantity that we accept from the i^{th} sell bid.
  - $v^{j,k}_d$, the quantity that will be stored in the reservoir at delivery time d in scenario j, k.

The intraday trading problem for this simple setting is described as follows.

$$
\begin{aligned}
(P): \max \sum_{d \in D} \Bigg(& \sum_{i \in I_{1,d}} (p^b_i q^b_i - p^s_i q^s_i) \\
& + \frac{1}{N_1} \sum_{j \in S_1} \Bigg(\sum_{i \in I^j_{2,d}} (p^b_i q^b_i - p^s_i q^s_i) \\
& + \frac{1}{N_2} \sum_{k \in S_2} \Bigg(\sum_{i \in I^{j,k}_{3,d}} (p^b_i q^b_i - p^s_i q^s_i) \Bigg) \Bigg) \Bigg) \qquad (1)
\end{aligned}
$$

$$
\begin{aligned}
& q^{s/b}_i \geq 0, \forall i \in I_{1,d}, d \in D && (\nu^{s/b}_i) \quad (2) \\
& q^{s/b}_i \geq 0, \forall i \in I^j_{2,d}, d \in D, j \in S_1 && (\nu^{s/b}_i) \quad (3) \\
& q^{s/b}_i \geq 0, \forall i \in I^{j,k}_{3,d}, d \in D, j \in S_1, k \in S_2 && (\nu^{s/b}_i) \quad (4) \\
& q^{s/b}_i \leq Q^{s/b}_i, \forall i \in I_{1,d}, d \in D && (\mu^{s/b}_i) \quad (5) \\
& q^{s/b}_i \leq Q^{s/b}_i, \forall i \in I^j_{2,d}, d \in D, j \in S_1 && (\mu^{s/b}_i) \quad (6) \\
& q^{s/b}_i \leq Q^{s/b}_i, \forall i \in I^{j,k}_{3,d}, d \in D, j \in S_1, k \in S_2 && (\mu^{s/b}_i) \quad (7) \\
& v^{j,k}_d = v^{j,k}_{d-1} + \sum_{i \in I_{1,d}} (q^s_i - q^b_i) \\
& \quad + \sum_{i \in I^j_{2,d}} (q^s_i - q^b_i) \\
& \quad + \sum_{i \in I^{j,k}_{3,d}} (q^s_i - q^b_i), \forall j \in S_1, k \in S_2, d \in D && (\lambda^{j,k}_d) \quad (8) \\
& 0 \leq v^{j,k}_d, \forall d \in D, \forall j \in S_1, \forall k \in S_2 && (\beta^{j,k}_d) \quad (9) \\
& v^{j,k}_d \leq V, \forall d \in D, \forall j \in S_1, \forall k \in S_2 && (\gamma^{j,k}_d) \quad (10)
\end{aligned}
$$

Objective function (1) is the profit of the agent, i.e. the difference between what is sold and what is bought. Constraints (2), (3), (4), (5), (6) and (7) state that we can only accept a non-negative quantity of the bids, and less than the maximal quantity of the bid. Constraint (8) is used in order to ensure the balance of the amount of energy that is stored in the reservoir. Constraints (9) and (10) ensure that the stored energy is within the reservoir limit for all possible scenarios.

The first order optimality condition with respect to the q variables can be written:

$$p_i^b - \sum_{j\in S_1}\sum_{k\in S_2}\lambda_d^{j,k} - \mu_i^b + \nu_i^b = 0, \forall i \in I_{1,d}, d \in D \quad (11)$$

$$-p_i^s + \sum_{j\in S_1}\sum_{k\in S_2}\lambda_d^{j,k} - \mu_i^b + \nu_i^b = 0, \forall i \in I_{1,d}, d \in D \quad (12)$$

$$\frac{1}{N_1}p_i^b - \sum_{k\in S_2}\lambda_d^{j,k} - \mu_i^b + \nu_i^b = 0, \forall i \in I_{2,d}^j, \quad \forall j \in S_1, d \in D \quad (13)$$

$$-\frac{1}{N_1}p_i^s + \sum_{k\in S_2}\lambda_d^{j,k} - \mu_i^b + \nu_i^b = 0, \forall i \in I_{2,d}^j, \quad \forall j \in S_1, d \in D \quad (14)$$

$$\frac{1}{N_1}\frac{1}{N_2}p_i^b - \lambda_d^{j,k} - \mu_i^b + \nu_i^b = 0, \forall i \in I_{3,d}^{j,k}, j \in S_1, \quad k \in S_2, d \in D \quad (15)$$

$$-\frac{1}{N_1}\frac{1}{N_2}p_i^s + \lambda_d^{j,k} - \mu_i^b + \nu_i^b = 0, \forall i \in I_{3,d}^{j,k}, j \in S_1, \quad k \in S_2, d \in D \quad (16)$$

From equations (11) and (12), we obtain

- If $p_i^b > \sum_{j\in S_1}\sum_{k\in S_2}\lambda_d^{j,k}$ then $\nu_i^b = 0$ and $\mu_i^b > 0$, which implies that $q_i^b = Q_i^b$.
- If $p_i^b < \sum_{j\in S_1}\sum_{k\in S_2}\lambda_d^{j,k}$ then $\mu_i^b = 0$ and $\nu_i^b > 0$, which implies that $q_i^b = 0$.
- If $p_i^b = \sum_{j\in S_1}\sum_{k\in S_2}\lambda_d^{j,k}$, then the bid is partially accepted.
- If $p_i^s < \sum_{j\in S_1}\sum_{k\in S_2}\lambda_d^{j,k}$ then $\nu_i^s = 0$ and $\mu_i^s > 0$, which implies that $q_i^s = Q_i^s$.
- If $p_i^s > \sum_{j\in S_1}\sum_{k\in S_2}\lambda_d^{j,k}$ then $\mu_i^s = 0$ and $\nu_i^s > 0$, which implies that $q_i^s = 0$.
- If $p_i^s = \sum_{j\in S_1}\sum_{k\in S_2}\lambda_d^{j,k}$, then the bid is partially accepted.

Therefore, it can be observed that, for the first time step, $\sum_{j\in S_1}\sum_{k\in S_2}\lambda_d^{j,k}$ can be interpreted as a threshold above which we accept sell offers, and below which we accept buy offers.

The same analysis can be performed for the second time step, using equations (13) and (14), and for the third time step using equations (15) and (16). For the second time step, we obtain that the threshold is $N_1\sum_{k\in S_2}\lambda_d^{j,k}$, and for the third time step, the threshold is $N_1N_2\lambda_d^{j,k}$.

These derivations demonstrate that a threshold policy is optimal for this simple setting. Although this is not a proof of the fact that threshold policies are optimal for multi-period generalizations of this simple model (such as the problem that is being considered in the paper), they are *indications* of the fact that a threshold policy is a reasonable starting point.

APPENDIX C
EXAMPLE OF ONE ITERATION OF THE REINFORCE ALGORITHM

In this example, we consider 3 time steps and 2 delivery periods. The settings are presented in figure 2. At the first two time steps, we can trade for both delivery times. At the last time step, we can only trade for the last delivery time, because the market for the first delivery time is closed.

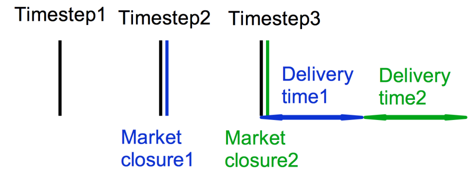

Fig. 2: Example settings for the response to question 3 of reviewer 1.

The data of the available bids is presented for the sell bids in TABLE I and for the buy bids in TABLE II. We assume that there is only one bid available for each delivery time at each time step. We also assume that the quantity of the different bids is equal to 50MWh, and that the initial volume in the reservoir is 0 MWh. We initialize the vector parameter θ which parametrizes our threshold policy according to the values that are provided in TABLE III. Finally, we consider 7 possible actions: (i) Buying 30 MWh, buying 20 MWh, buying 10 MWh, no trade, selling 10 MWh, selling 20 MWh, selling 30 MWh. We assume that our step size γ is equal to 10^{-4}.

Timestep/delivery time	1	2
1	20	29
2	18	26
3	/	30

TABLE I: Price of the sell bids.

Timestep/delivery time	1	2
1	14	22
2	16	24
3	/	28

TABLE II: Price of the buy bids.

Parameter	Value
μ_X	20
μ_Y	25
$\exp(\sigma_X)$	1.3
$\exp(\sigma_Y)$	0.8

TABLE III: Value of the elements of the parameter vector θ

The algorithm has two phases: (a) we run an episode based on the vector parameter θ; (b) we update the parameter vector θ based on the results of the episode.

a) Run the episode: At the first time step, we draw the buy thresholds $X_1, X_2 \sim \mathbb{N}(20, 1.3)$ and the sell thresholds $Y_1, Y_2 \sim \mathbb{N}(25, 0.8)$. Suppose that we obtain $X_1 = 19.83\frac{€}{MWh}$, $X_2 = 20.24\frac{€}{MWh}$, $Y_1 = 24.32\frac{€}{MWh}$, $Y_2 = 26.44\frac{€}{MWh}$. At this time step, (i) we do not buy anything for the first delivery hour because the buy threshold ($19.83\frac{€}{MWh}$) is lower than the most advantageous sell bid ($20\frac{€}{MWh}$); (ii) we do not buy anything for the second delivery hour because the buy threshold ($20.24\frac{€}{MWh}$) is lower than the most advantageous sell bid ($29\frac{€}{MWh}$); (iii) we do not sell anything for the first delivery hour because the sell threshold ($24.32\frac{€}{MWh}$) is higher than the most advantageous buy bid ($14\frac{€}{MWh}$); (iv) we do not sell anything for the second delivery hour because the sell threshold ($26.44\frac{€}{MWh}$) is higher than the most advantageous buy bid ($22\frac{€}{MWh}$).

At the second time step, we have to draw the buy thresholds $X_1, X_2 \sim \mathbb{N}(20, 1.3)$ and the sell thresholds $Y_1, Y_2 \sim \mathbb{N}(25, 0.8)$. Suppose that we obtain $X_1 = 21.33\frac{€}{MWh}$, $X_2 = 18.24\frac{€}{MWh}$, $Y_1 = 26.14\frac{€}{MWh}$, $Y_2 = 24.84\frac{€}{MWh}$. At this time step, we accept to buy the biggest possible action 30MWh for the first delivery time because the buy threshold ($21.33\frac{€}{MWh}$) is higher than the price for buying 25MWh (25MWh is the midpoint between 20MWh and 30MWh) which is $18\frac{€}{MWh}$.

At the third time step, we only draw the thresholds for the second delivery hour because the product of the first delivery hour is closed. Suppose that we obtain $X_2 = 22.47\frac{€}{MWh}$, $Y_2 = 26.37\frac{€}{MWh}$. At this time step, we accept to sell the greatest possible quantity 30MWh at the second delivery time, because the sell threshold ($26.37\frac{€}{MWh}$) is lower than the price for selling 25 MWh ($28\frac{€}{MWh}$).

When we finish running the episode, we can collect the different elements which are needed in order to update the parameter vector. We need to store 3 elements:

1) **The visited states:** As the market price is the only information of the state that is used in order to apply our decision, we only need to store that part of the state.

$$
\begin{aligned}
&(p_{1,1}(-25), p_{1,2}(-25), p_{2,1}(-25), p_{2,2}(-25), p_{3,2}(-25)) \\
&= (20, 29, 18, 26, 30) \\
&(p_{1,1}(-15), p_{1,2}(-15), p_{2,1}(-15), p_{2,2}(-15), p_{3,2}(-15)) \\
&= (20, 29, 18, 26, 30) \\
&(p_{1,1}(-5), p_{1,2}(-5), p_{2,1}(-5), p_{2,2}(-5), p_{3,2}(-5)) \\
&= (20, 29, 18, 26, 30) \\
&(p_{1,1}(5), p_{1,2}(5), p_{2,1}(5), p_{2,2}(5), p_{3,2}(5)) \\
&= (14, 22, 16, 24, 28) \\
&(p_{1,1}(15), p_{1,2}(15), p_{2,1}(15), p_{2,2}(15), p_{3,2}(15)) \\
&= (14, 22, 16, 24, 28) \\
&(p_{1,1}(25), p_{1,2}(25), p_{2,1}(25), p_{2,2}(25), p_{3,2}(25)) \\
&= (14, 22, 16, 24, 28)
\end{aligned}
$$

2) **The chosen action:** The different actions that we have taken are $(a_{1,1}, a_{1,2}, a_{2,1}, a_{2,2}, a_{3,2}) = (0, 0, -30, 0, 30)$
3) **The observed cumulative reward:** We observe 1 reward per step: $(r_1, r_2, r_3) = (0, -540, 840)$. The cumulative reward is therefore given by: $(g_1, g_2, g_3) = (300, 300, 840)$.

b) Update the threshold: We update the threshold using the REINFORCE algorithm.

$$
\begin{aligned}
\mu_X \leftarrow \mu_X + \gamma\Big(&\frac{\phi(p_{1,1}(-5); \mu_X, \exp(\sigma_X))}{1 - \Phi(p_{1,1}(-5); \mu_X, \exp(\sigma_X))} g_1 \\
&+ \frac{\phi(p_{1,2}(-5); \mu_X, \exp(\sigma_X))}{1 - \Phi(p_{1,2}(-5); \mu_X, \exp(\sigma_X))} g_1 \\
&+ \frac{-\phi(p_{2,1}(-25); \mu_X, \exp(\sigma_X))}{\Phi(p_{2,1}(-25); \mu_X, \exp(\sigma_X))} g_2 \\
&+ \frac{\phi(p_{2,2}(-5); \mu_X, \exp(\sigma_X))}{1 - \Phi(p_{2,2}(-5); \mu_X, \exp(\sigma_X))} g_2 \\
&+ \frac{\phi(p_{3,2}(-5); \mu_X, \exp(\sigma_X))}{1 - \Phi(p_{3,2}(-5); \mu_X, \exp(\sigma_X))} g_3\Big) \\
&= 20 + 10^{-4} \cdot \Big(\frac{\phi(20; 20, 1.3)}{1 - \Phi(20; 20, 1.3)} 300 \\
&+ \frac{\phi(29; 20, 1.3)}{1 - \Phi(29; 20, 1.3)} 300 + \frac{-\phi(18; 20, 1.3)}{\Phi(18; 20, 1.3)} 300 \\
&+ \frac{\phi(26; 20, 1.3)}{1 - \Phi(26; 20, 1.3)} 300 + \frac{\phi(30; 20, 1.3)}{1 - \Phi(30; 20, 1.3)} 840\Big) \\
&= 20.73
\end{aligned}
$$

$$
\begin{aligned}
\mu_Y \leftarrow \mu_Y + \gamma\Big(&\frac{-\phi(p_{1,1}(5); \mu_Y, \exp(\sigma_Y))}{\Phi(p_{1,1}(5); \mu_Y, \exp(\sigma_Y))} g_1 \\
&+ \frac{-\phi(p_{1,2}(5); \mu_Y, \exp(\sigma_Y))}{\Phi(p_{1,2}(5); \mu_Y, \exp(\sigma_Y))} g_1 \\
&+ \frac{-\phi(p_{2,1}(5); \mu_Y, \exp(\sigma_Y))}{\Phi(p_{2,1}(5); \mu_Y, \exp(\sigma_Y))} g_2 \\
&+ \frac{-\phi(p_{2,2}(5); \mu_Y, \exp(\sigma_Y))}{\Phi(p_{2,2}(5); \mu_Y, \exp(\sigma_Y))} g_2 \\
&+ \frac{\phi(p_{3,2}(25); \mu_Y, \exp(\sigma_Y))}{1 - \Phi(p_{3,2}(25); \mu_Y, \exp(\sigma_Y))} g_3\Big) \\
&= 25 + 10^{-4} \cdot \Big(\frac{-\phi(14; 25, 0.8)}{\Phi(14; 25, 0.8)} 300 \\
&+ \frac{-\phi(22; 25, 0.8)}{\Phi(22; 25, 0.8)} 300 + \frac{-\phi(16; 25, 0.8)}{\Phi(16; 25, 0.8)} 300 \\
&+ \frac{-\phi(24; 25, 0.8)}{\Phi(24; 25, 0.8)} 300 + \frac{\phi(28; 25, 0.8)}{1 - \Phi(28; 25, 0.8)} 840\Big) \\
&= 24.24
\end{aligned}
$$

The development for the standard deviation is similar. After updating the parameter vector θ, we can start a new iteration by running a new episode. It can be observed in this example that we only use the price available in the CIM, and that we do not make assumption on their future evolution.

Appendix D
Relative value of observable bids

In this appendix, we describe how to compute the closed-form expression of $\pi_\theta^d(a|s)$ and its derivative for the different actions if the rolling intrinsic method sells 20 MWh for delivery hour d. The graphical illustration of these probabilities is shown in Fig. 3. In this situation, we have 4 possible actions: sell 0, 10, 20 or 30 MWh. The computation of the closed-form solution for 0 and 10 MWh is exactly the same as in the initial situation, as shown in Eqs. (17) and (18). The reason is that these actions are feasible, and do not corresponds to the feasible action with the greatest volume of trading (*Sell* 20 *MWh*), therefore we wish to keep their probability of being selected intact. For the action *Sell* 30 *MWh*, the development is different. As explained in the paper, as rolling intrinsic accepts less than 30 MWh for that action, the threshold is drawn from the auxiliary Gaussian distribution. The mathematical development is shown in equation 20. The difference with the initial situation is that the auxiliary Gaussian distribution has a higher mean, which implies that the probability of this action is decreased. For the action *Sell* 20 *MWh*, the development has also changed because it is the action with the greatest amount of traded volume that is smaller than the quantity accepted by rolling intrinsic. Therefore, for this action we allocate the probability mass that has been removed from the action *Sell* 30 *MWh*. The mathematical development is shown in equation 19.

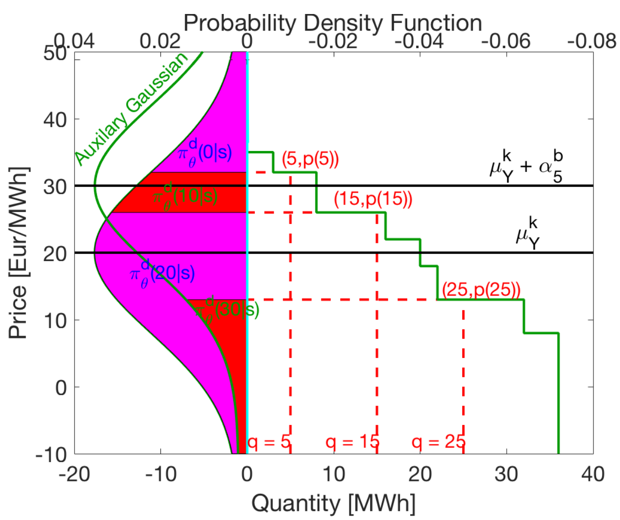

Fig. 3: Illustration of the probability reallocation that relies on the auxiliary Gaussian distribution.

- **Sell** 0 **MWh:**

$$\begin{aligned}\pi_\theta^d(0|s) &= \Pr(a_{t,d}=0)\\ &= \Pr(Y_d \geq p(5))\\ &= 1-\Pr(Y_d \leq p(5))\\ &= 1-\Phi(p(5);\mu_Y^k,\exp(\sigma_Y))\end{aligned} \tag{17}$$

- **Sell** 10 **MWh:**

$$\begin{aligned}\pi_\theta^d(10|s) &= \Pr(a_{t,d}=10)\\ &= \Pr(p(15)\leq Y_d \leq p(5))\\ &= \Pr(Y_d \leq p(5)) - \Pr(Y_d \leq p(15))\\ &= \Phi(p(5);\mu_Y^k,\exp(\sigma_Y))\\ &\quad - \Phi(p(15);\mu_Y^k,\exp(\sigma_Y))\end{aligned} \tag{18}$$

- **Sell** 20 **MWh:**

$$\begin{aligned}\pi_\theta^d(20|s) &= \Pr(a_{t,d}=20)\\ &= \Pr(Y_d \leq p(15)) - \Pr(Y_d+\alpha_5^b \leq p(25))\\ &= \Phi(p(15);\mu_Y^k,\exp(\sigma_Y))\\ &\quad - \Phi(p(25);\mu_Y^k+\alpha_5^b,\exp(\sigma_Y))\end{aligned} \tag{19}$$

- **Sell** 30 **MWh:**

$$\begin{aligned}\pi_\theta^d(30|s) &= \Pr(a_{t,d}=30)\\ &= \Pr(Y_d+\alpha_5^b \leq p(25))\\ &= \Phi(p(25);\mu_Y^k+\alpha_5^b,\exp(\sigma_Y))\end{aligned} \tag{20}$$

In order to implement the REINFORCE algorithm, we also need the policy derivative for the different actions. The development in order to compute these is made in Eqs. (21), (22), (23) and (24). The derivatives for 0 and 10 MWh are equal to 0 because α_5^b does not appear in the probability of these actions. The derivatives for the action *Sell* 30 *MWh* is negative. This is coherent with the intuition that the higher the value of α_5^b, the less likely we are to choose the action that is not selected by rolling intrinsic. The derivatives for the action *Sell* 20 *MWh* is the opposite of the one of the action *Sell* 30 *MWh*. This is due to the fact that the probability lost by the action *Sell* 30 *MWh* is exactly transferred to the action *Sell* 20 *MWh*.

$$\frac{\partial \pi_\theta^d(0|s)}{\partial \alpha_5^b} = 0 \tag{21}$$

$$\frac{\partial \pi_\theta^d(10|s)}{\partial \alpha_5^b} = 0 \tag{22}$$

$$\frac{\partial \pi_\theta^d(20|s)}{\partial \alpha_5^b} = \phi(p(25),\mu_Y^k+\alpha_5^b,\exp(\sigma_Y)) \tag{23}$$

$$\frac{\partial \pi_\theta^d(30|s)}{\partial \alpha_5^b} = -\phi(p(25),\mu_Y^k+\alpha_5^b,\exp(\sigma_Y)) \tag{24}$$

Appendix E
Rolling intrinsic model with the round-trip efficiency

In this appendix, we present the optimization model that the rolling intrinsic method solves at each time step $t \in T$. We start by defining the notation. Then we present the model.

- The sets are:
  - T, the set of time steps.
  - D, the set of delivery times.
  - $I_{t,d}, \forall t \in T, d \in D$, the set of bids that are available for a given time step and delivery time.
- The parameters are:
  - $p_i^b, i \in I_{t,d}$, the price of the i^{th} buy bid at time step t and delivery time d.
  - $p_i^s, i \in I_{t,d}$, the price of the i^{th} sell bid at time step t and delivery time d.
  - $Q_i^b, i \in I_{t,d}$, the available quantity of the i^{th} buy bid at time step t and delivery time d.
  - $Q_i^s, i \in I_{t,d}$, the available quantity of the i^{th} sell bid at time step t and delivery time d.

- $F_{t,d}^{s,r}$, the quantity which has been sold but not yet bought back before time step t for delivery hour d. Mathematically, it can be computed as: $F_{t,d}^{s,r} = \max\left(\sum_{\tau<t}\sum_{i\in I_{\tau,d}}\left(q_i^b - q_i^s\right), 0\right)$
- $F_{t,d}^{b,r}$, the quantity which has been bought and not yet sold back before time step t for delivery hour d. Mathematically, it can be computed as: $F_{t,d}^{b,r} = \max\left(\sum_{\tau<t}\sum_{i\in I_{\tau,d}}\left(q_i^s - q_i^b\right), 0\right)$
- V, the maximum capacity of the reservoir.
- η_{in}, the charging efficiency.
- η_{out}, the discharging efficiency.

- The variables are:
  - $q_i^{b,r}, i \in I_{t,d}$, the quantity that we accept from the i^{th} buy bid for delivery time d at time step t.
  - $q_i^s, i \in I_{t,d}$, the quantity that we accept from the i^{th} sell bid for delivery time d at time step t.
  - $v_{t,d}$, the quantity that would be stored in the reservoir at delivery time d, if we were only executing the trade realized at time step t or before.
  - $f_{t,d}^{b,r}$, the quantity sold for delivery hour d at time step t which is financially covered. This means that this trade is canceling a quantity that we have bought at previous time steps for delivery hour d.
  - $f_{t,d}^{b,n}$, the quantity sold for delivery hour d at time step t which is physically covered. This means that this trade is not canceling a quantity that we have bought at previous time steps for delivery hour d.
  - $f_{t,d}^{s,r}$, the quantity bought for delivery hour d at time step t which is financially covered. This means that this trade is canceling a quantity that we have sold at previous time steps for delivery hour d.
  - $f_{t,d}^{s,n}$, the quantity bought for delivery hour d at time step t which is physically covered. This means that this trade is not canceling a quantity that we have sold at previous time steps for delivery hour d.
  - $x_{t,d}^{b,r}$, a commitment variable which indicates if we are selling power for delivery hour d at time step t.
  - $x_{t,d}^{b,n}$, a commitment variable which indicates if we are selling more power than the quantity bought before time step t for delivery hour d.
  - $x_{t,d}^{s,r}$, a commitment variable which indicates if we are buying power for delivery hour d at time step t.
  - $x_{t,d}^{s,n}$, a commitment variable which indicates if we are buying more power than the quantity bought before time step t for delivery hour d.

Objective (25) determines the profit as the sum of the revenue for what we sell minus the cost of what we buy. Constraint (26) requires that we cannot accept a quantity greater than the maximum capacity of the bid. Constraint (27) states that (i) we cannot financially trade more than the quantity sold/bought at earlier time steps (ii) in order to buy/sell, the associated binary variable should be activated. Constraint (28) represents the fact that we cannot physically trade if the associated binary variable is not activated. Constraints (29) and (30) imply that, in order to physically (buy/sell) power, we need to have completely canceled the previous position by financially trading. Constraint (31) states that we cannot buy and sell power for the same delivery time. Constraint (32) links the quantity accepted for each bid and the total quantity exchanged for the delivery hour. Constraint (33) shows how the capacity of the reservoir evolves with respect to the quantity that we trade. Constraint (34) requires that the capacity stored in the reservoir should stay in the reservoir limit.

$$(P_t): \quad \max \sum_{d\in D}\sum_{i\in I_{t,d}}\left(p_i^b q_i^b - p_i^s q_i^s\right) \tag{25}$$

$$0 \le q_i^{s/b} \le Q_i^{s/b}, \forall i \in I_{t,d}, \forall d \in D \tag{26}$$

$$0 \le f_{t,d}^{s/b,r} \le F_{t,d}^{s/b,r} x_{t,d}^{s/b,r}, \forall d \in D \tag{27}$$

$$0 \le f_{t,d}^{s,n} \le \frac{V}{\eta_{\text{in}}} x_{t,d}^{s/b,n}, \forall d \in D \tag{28}$$

$$x_{t,d}^{s/b,n} \cdot F_{t,d}^{s/b,r} \le f_{t,d}^{s/b,r}, \forall d \in D \tag{29}$$

$$x_{t,d}^{s/b,n} \le x_{t,d}^{s/b,r}, \forall d \in D \tag{30}$$

$$x_{t,d}^{s,r} + x_{t,d}^{b,r} \le 1, \forall d \in D \tag{31}$$

$$\sum_{i\in I_{t,d}}\left(q_i^s - q_i^b\right) = f_{t,d}^{s,r} + f_{t,d}^{s,n} - f_{t,d}^{b,r} - f_{t,d}^{b,n}, \forall d \in D \tag{32}$$

$$\begin{aligned} v_{t,d} &= v_{t-1,d} \\ &\quad + \sum_{a\in D|a\le d}\Big(\eta_{\text{in}} f_{t,a}^{s,n} + (1/\eta_{\text{out}}) f_{t,a}^{s,r} \\ &\quad - (1/\eta_{\text{out}}) f_{t,a}^{b,n} - \eta_{\text{in}} f_{t,a}^{b,r}\Big), \forall d \in D \end{aligned} \tag{33}$$

$$0 \le v_{t,d} \le V, \forall d \in D \tag{34}$$

Appendix F
Computational scalability

In this appendix, we analyse the computational scalability of our learning method with respect to the trading frequency. The results are presented in figure 4. The figure presents the run time for simulating the 200 first days of 2015 at different trading frequencies. These simulations have been realized 10 times in order to obtain more reliable estimates of run times, and the average run time is presented on the graph[1]. We observe that the run time is proportional to the time steps that are used in the training, which is expected, since the number of operations in the REINFORCE algorithm scales linearly with the number of training samples that we employ. Note a fixed run time which is required for the market simulation, and which is independent of the number of time steps.

Appendix G
Learning process stability

In order to test the stability of the learning algorithm, we have run 6 different realizations of the REINFORCE algorithm at hourly frequency. We have executed 6 different realizations of the REINFORCE algorithm at hourly frequency (recall that REINFORCE is a stochastic gradient algorithm). Each of these simulations has been performed on an HPC cluster, where we

[1]This experiment has been realized in parallel on 3 cores (2.5 GHz Intel Core i7).

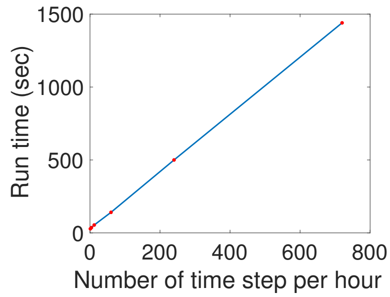

Fig. 4: Evolution of the run time for 200 episodes with respect to the trading frequency.

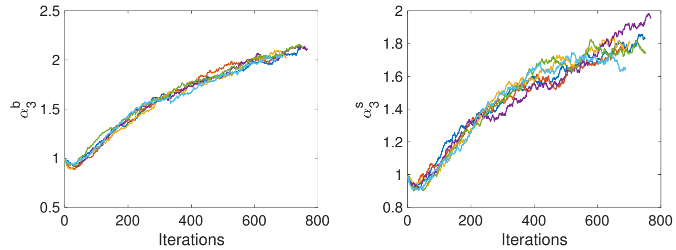

Fig. 7: Evolution of α_3^b(left) and α_3^s(right) for 6 realizations of the REINFORCE algorithm.

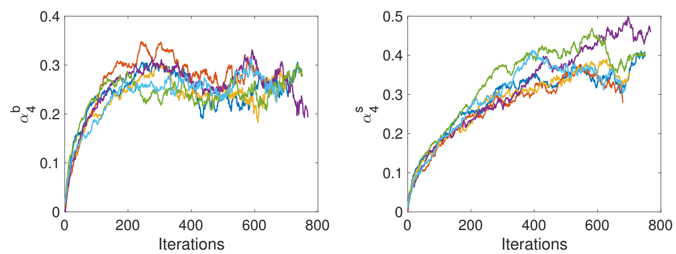

Fig. 8: Evolution of α_4^b(left) and α_4^s(right) for 6 realizations of the REINFORCE algorithm.

have used 8 CPUs for 8 hours. The evolution of the parameters is presented in figures 5, 6, 7, 8 and 9. These graphs suggest that the parameter evolution is very similar for the 6 different realizations of the algorithm. This can also be observed for the evolution of the profit, which is presented in figure 10. This implies that, even if the learning algorithm is stochastic, its evolution is stable. Note that the alpha parameters that have the lowest influence on the threshold also present the greatest variation among the 6 simulations.

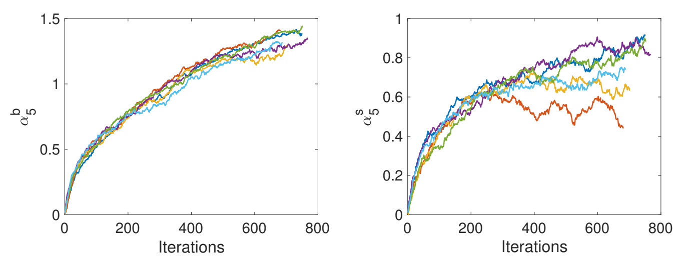

Fig. 9: Evolution of α_5^b(left) and α_5^s(right) for 6 realizations of the REINFORCE algorithm.

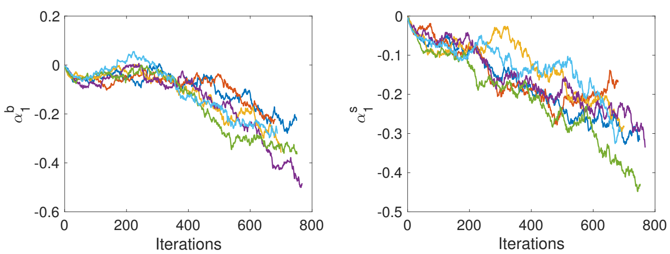

Fig. 5: Evolution of α_1^b(left) and α_1^s(right) for 6 realizations of the REINFORCE algorithm.

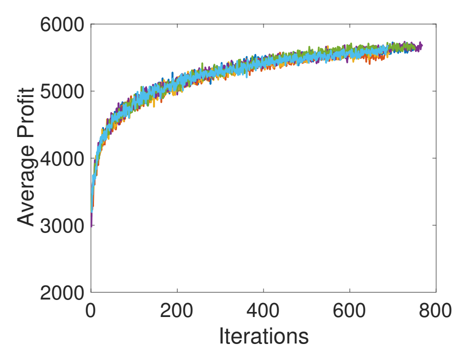

Fig. 10: Evolution of the daily profit for 6 realizations of the REINFORCE algorithm.

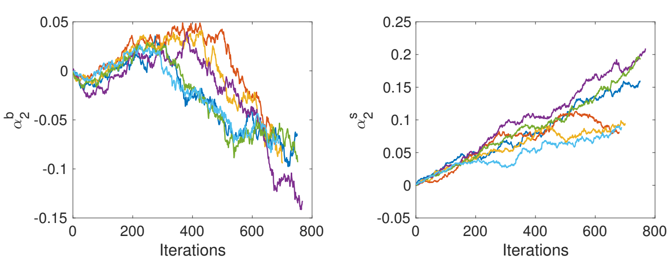

Fig. 6: Evolution of α_2^b(left) and α_2^s(right) for 6 realizations of the REINFORCE algorithm.

Appendix H
Explanation of the threshold policy graph step by step

We will explain step by step all the elements on the graph. This graph illustrates the threshold policy for the hydro problem with the following settings:

- We consider four possible actions: sell 0, 10, 20 or 30 MWh.
- The buy bids available on the market are presented in table IV
- $\mu_Y = 20$
- $\exp(\sigma_Y) = 12$

We now illustrate, step by step, the construction of the graph:

1) We draw the demand curve that is illustrated in table

Price (Eur/MWh)	Quantity (MWh)
35	3
32	5
26	8
22	4
18	2
13	10
8	4

TABLE IV: Buy bids available on the market.

IV, which corresponds to the buy bids available on the market. We obtain figure 11 (left). The value of the demand curve should be read on the lower x-axis.

2) We draw the Gaussian distribution with mean μ_Y and standard deviation $\exp(\sigma_Y)$. This bell curve indicates the probability density function of the sell threshold. We obtain figure 11 (right). The value of the Bell curve should be read on the upper x-axis.
3) We take the image of the mid points between the different actions (0MWH, 10MWh, 20MWh, 30MWh). In our case, it is 5MWh which is the midpoint between 0 and 10 MWh, while 15MWh which is the midpoint between 10 and 20 MWh and 25MWh which is the midpoint between 20 and 30 MWh. We obtain figure 12 (left).
4) Finally, we know that the probability of each action can be expressed in closed form as:

$$\pi^d_\theta(0|s) = \Pr(p(5) \leq Y_d)$$
$$\pi^d_\theta(10|s) = \Pr(p(15) \leq Y_d \leq p(5))$$
$$\pi^d_\theta(20|s) = \Pr(p(25) \leq Y_d \leq p(15))$$
$$\pi^d_\theta(30|s) = \Pr(Y_d \leq p(25))$$

Therefore, in order to observe these probabilities, we need to determine where the image of the midpoint intersects the bell curve. This is shown in the final figure 12 (right). The probability of each of the four actions is indicated by the two purple segments and the two red segments of the bell curve.

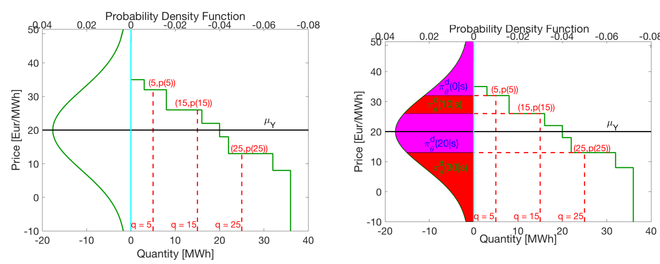

Fig. 12: Threshold policy graph construction step 3 (left) and step 4 (right).

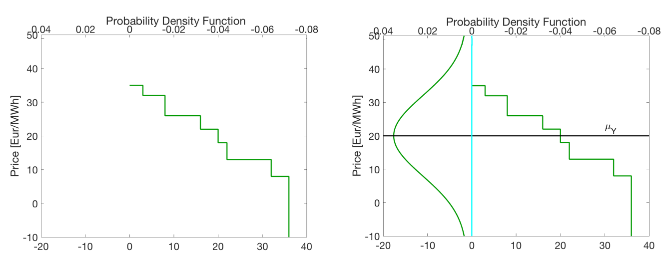

Fig. 11: Threshold policy graph construction step 1 (left) and step 2 (right).